Thursday, February 27, 2025

Section 3.7

LEC20 - Improper Integrals Continued

Warm-up Problem

Which of the following integrals are improper?

1. $\underbrace{\int_{\infty}^{-3} \frac{4}{\sqrt{1-x}}\,dx}_{\text{VA at } x=1}$ 2. $\underbrace{\int_{-2}^{8} \frac{1}{x-3}\,dx}_{\text{VA at } x=3}$ 3. $\int_{-4}^{4} \frac{1}{x^2+9}\,dx$ 4. $\int_{0}^{1} \frac{1}{x-5}\,dx$

Improper Integrals - Discontinuity at one end of the interval

If f is continuous on $[a,b)$ then $\int_a^b f(x)\,dx = \lim_{t\to b^-} \int_a^t f(x)\,dx$. In this case, we say the integral converges. If the limit does not exist, the integral diverges.

Analogously, if f is continuous on $(a,b]$, then $\int_a^b f(x)\,dx = \lim_{t\to a^+} \int_t^b f(x)\,dx$

Example

Does the following improper converge or diverge. If it converges, evaluate it.

$$\int_0^3 \frac{8}{\sqrt{6-2x}}\,dx = \lim_{t\to 3^-}\left[\int_0^t \frac{8}{\sqrt{6-2x}}\,dx\right] = \lim_{t\to 3^-}\left[\int_6^{6-2t} \frac{8}{\sqrt{u}} \cdot -\frac{1}{2}\,du\right] = -\frac{1}{2}\lim_{t\to 3^-}\left[16\sqrt{u}\Big|_6^{6-2t}\right] = -\frac{1}{2}\lim_{t\to 3^-}\left[16\sqrt{6-2t} - 16\sqrt{6}\right] = 8\sqrt{6}$$

Therefore, the improper integral converges to $8\sqrt{6}$

Example

Is the following computation? Can you tell from the final answer alone?

$$\underbrace{\int_{-1}^{3} \frac{1}{x^2}\,dx}_{\text{Discontinuous at } x=0\text{, so can't use FTC2}} = \int_{-1}^{3} x^{-2} = \left[-x^{-1}\right]_{-1}^{3} = -3^{-1} - 1 = -4/3 \leftarrow \text{Answer is negative but } f(x) \text{ always positive}$$

Improper Integrals - Discontinuity inside the interval

If f is continuous on $[a,b]$ except at a point c in (a,b), then $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$, provided that both of the integrals on the RHS converge. In this case, $\int_a^b f(x)\,dx$ converges. If either one of the integrals on the RHS diverge, then $\int_a^b f(x)\,dx$ diverges.

Example

Does the following improper integral converge or diverge? If it converges, evaluate it.

$$\int_{-1}^{3} \frac{1}{x^2}\,dx = \int_{-1}^{0} \frac{1}{x^2}\,dx + \int_{0}^{3} \frac{1}{x^2}\,dx = \lim_{t\to 0^-}\left[\int_{-1}^{t} \frac{1}{x^2}\,dx\right] + \lim_{z\to 0^+}\left[\int_{z}^{3} \frac{1}{x^2}\,dx\right] \leftarrow \text{Entire limit must therefore diverge}$$

$$= \lim_{t\to 0^-}\left[\left[-x^{-1}\right]_{-1}^{t}\right]$$

$$= \lim_{t\to 0^-}\left[-\frac{1}{t} + 1\right] \leftarrow \text{Diverges}$$

Improper Integrals - Integrating over $(-\infty, \infty)$

If f is continuous on $\mathbb{R}$, then $\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{\infty} f(x)\,dx$, provided that both of the integrals on the RHS converge. In this case, $\int_{-\infty}^{\infty} f(x)\,dx$ converges.

If either one of the integrals on the RHS diverges, then $\int_{-\infty}^{\infty} f(x)\,dx$ diverges.

Example

In statistics, the probability density function of a random variable X is a non-negative continuous function, p, that satisfies the following equation: $\int_{-\infty}^{\infty} p(x)\,dx = 1$. The probability that X lies between a and b is $\int_a^b p(x)\,dx$.

Find a value for the constant k such that $p(x) = \frac{k}{x^2+1}$ is a probability density function.

$$\int_{-\infty}^{\infty} p(x)\,dx = \int_{0}^{\infty} p(x)\,dx \quad + \quad \int_{-\infty}^{0} p(x)\,dx$$

$$= \lim_{t\to\infty}\left[\int_0^t p(x)\,dx\right] \quad = \quad \lim_{z\to -\infty}\left[\int_z^0 p(x)\,dx\right]$$

$$= k\lim_{t\to\infty}\left[\arctan(t) - \arctan(0)\right] + k\lim_{z\to-\infty}\left[\arctan(z) - \arctan(0)\right]$$

$$= k\pi \Rightarrow k\pi = 1 \Rightarrow k = \frac{1}{\pi}$$

LEC21 - Summary of Integration

Warm-Up Problem

What technique would work best to integrate $\int x^2 \sin(5x^3)\,dx$

$\Rightarrow$ looks like result of chain rule $\Rightarrow$ u-substitution

Examples

Which method would you use? (You don't need to evaluate the integrals)

1. $\int \frac{\cos x}{\sin^4 x}\,dx$ u-substitution
2. $\int t^3 \ln t\,dt$ Integration by parts
3. $\int \sec(3x)\tan(3x)$ write in terms of sin & cos
4. $\int \frac{\tan(\ln(x))}{x}\,dx$ u-substitution
5. $\int (z^2+3z)^2\,dz$ Expand
6. $\int 5x e^{6x}\,dx$ Integration by parts
7. $\int \frac{2x^6-1}{2x^7-7x}\,dx$ u-substitution
8. $\int \frac{x+2}{x(x^2-9)}\,dx$ Partial fractions
9. $\int x^3 \sin(3x^2)\,dx$ Integration by parts
10. $\int \frac{e^{5x}}{e^{2x}-1}\,dx$ Partial fractions (let $z = e^x$)

Choosing an Integration Method

1. Try easy things first
   - Expanding brackets/simplifying
   - Is it a known antiderivative
2. Try substitution if...
   - You see one function composed inside something else, and its derivative a multiplicative factor (a clear substitution)
3. Try longer methods
   - If its a rational function, try partial fractions
   - If it's a product of sines and cosines, try the techniques from 3.2
   - If its a product of two functions, or a single inverse, try integration by parts
4. Other methods
   - Try unusual substitutions

A few things to remember

1. When you have finished rewriting an integral, look through the steps again
   - You may need 2+ methods
   - You may notice something new
2. When practicing integration
   - Practice each method separately
   - Practice with a mix of questions
3. You can easily check your own antiderivatives
   - Differentiate the antiderivative
4. Do not make up your own integral problems
   - Integration is harder than differentiation
   - Many integrals are impossible to solve